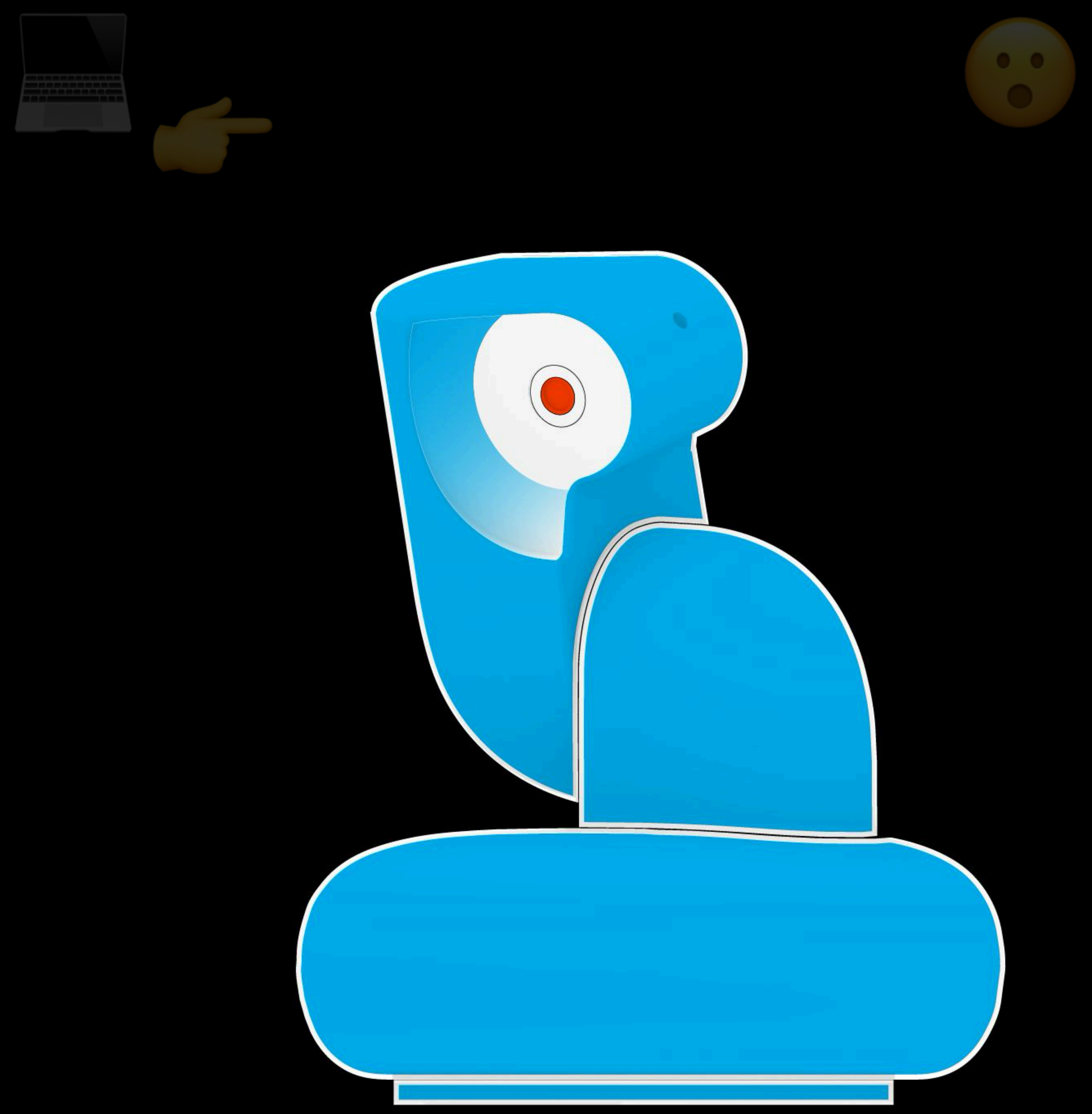
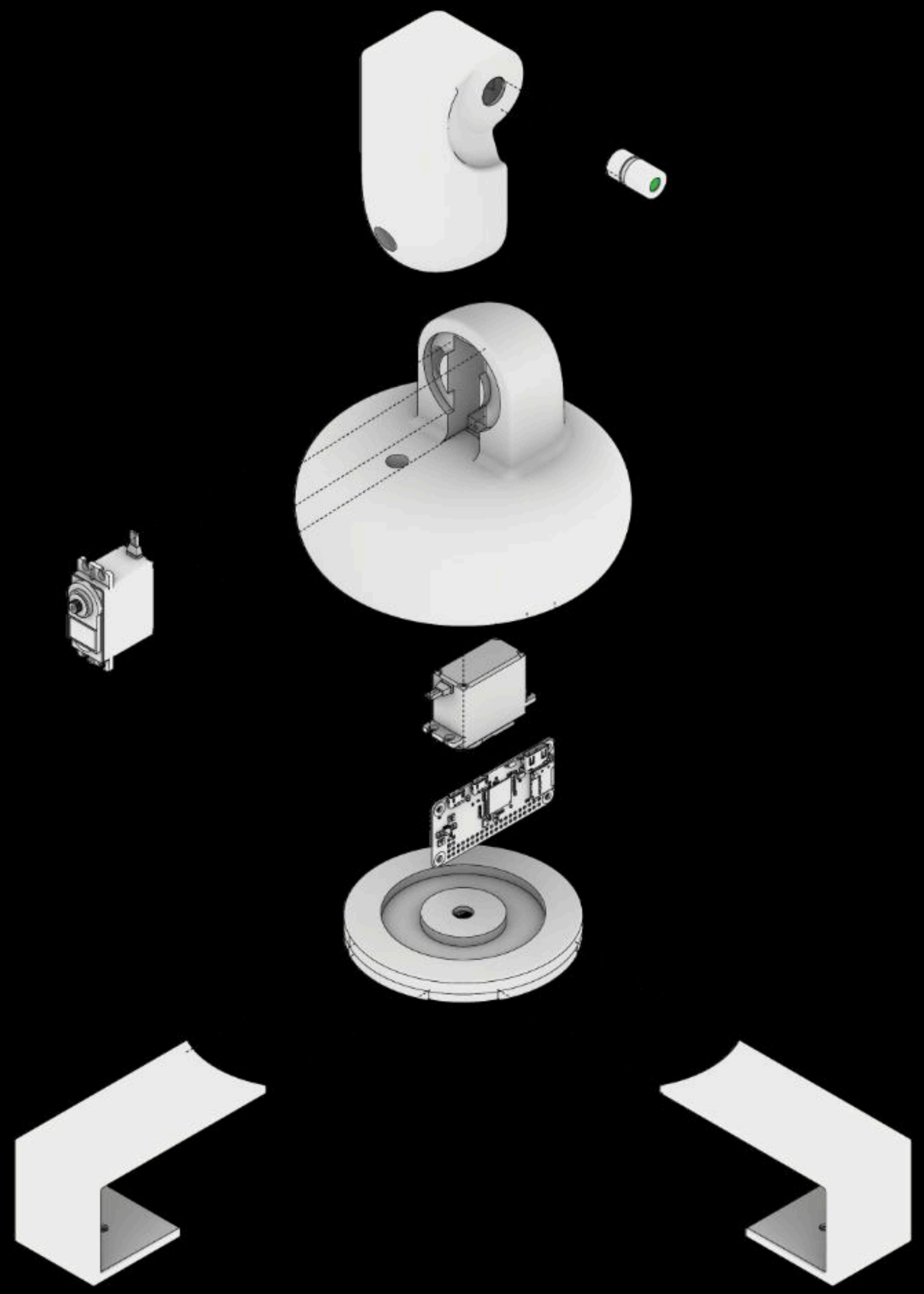
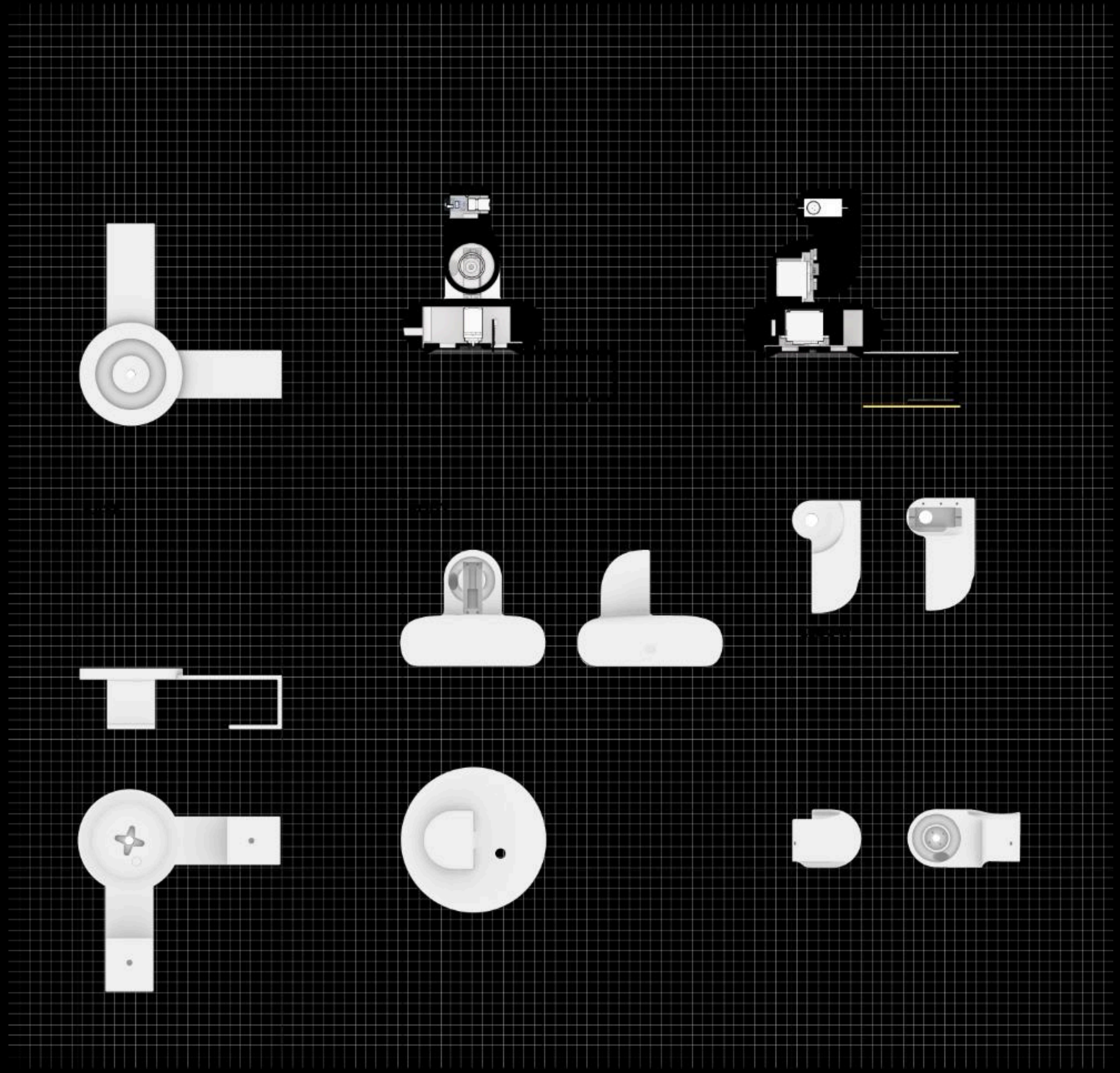


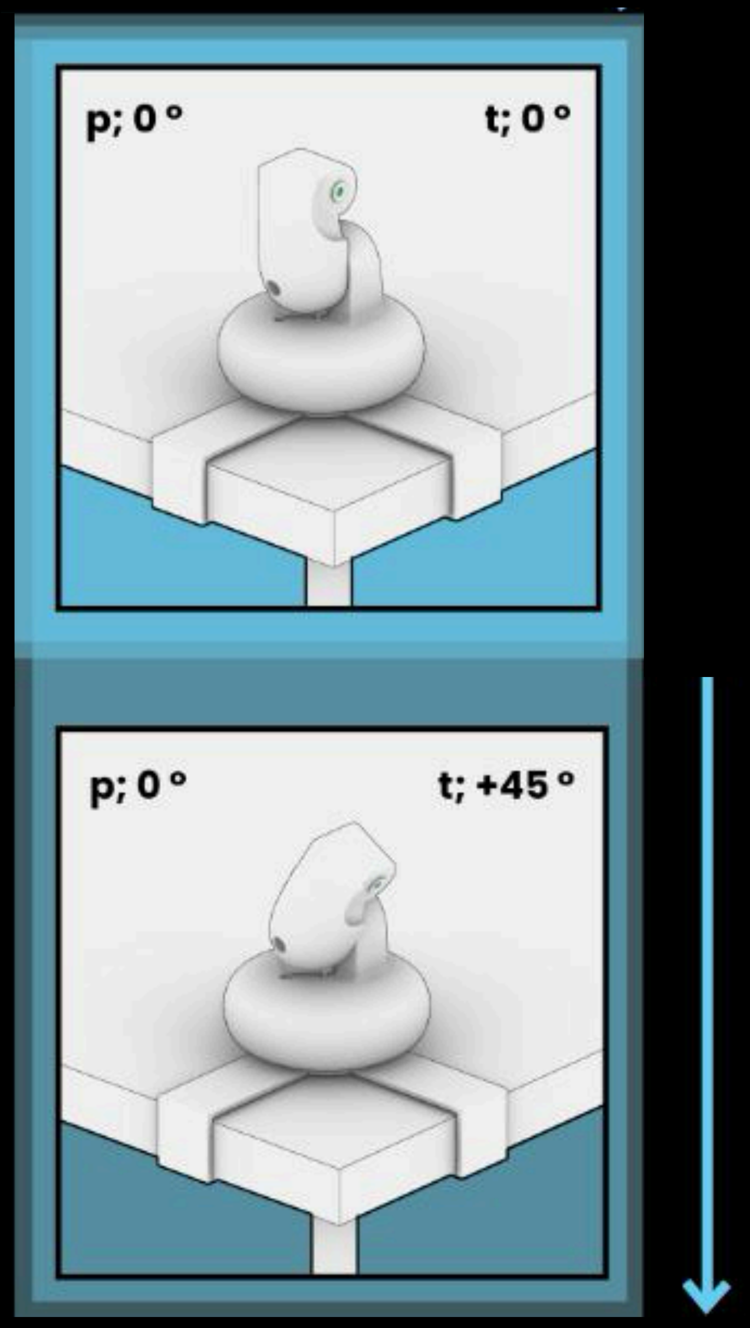
DEMO TIME!

Inside Pointo



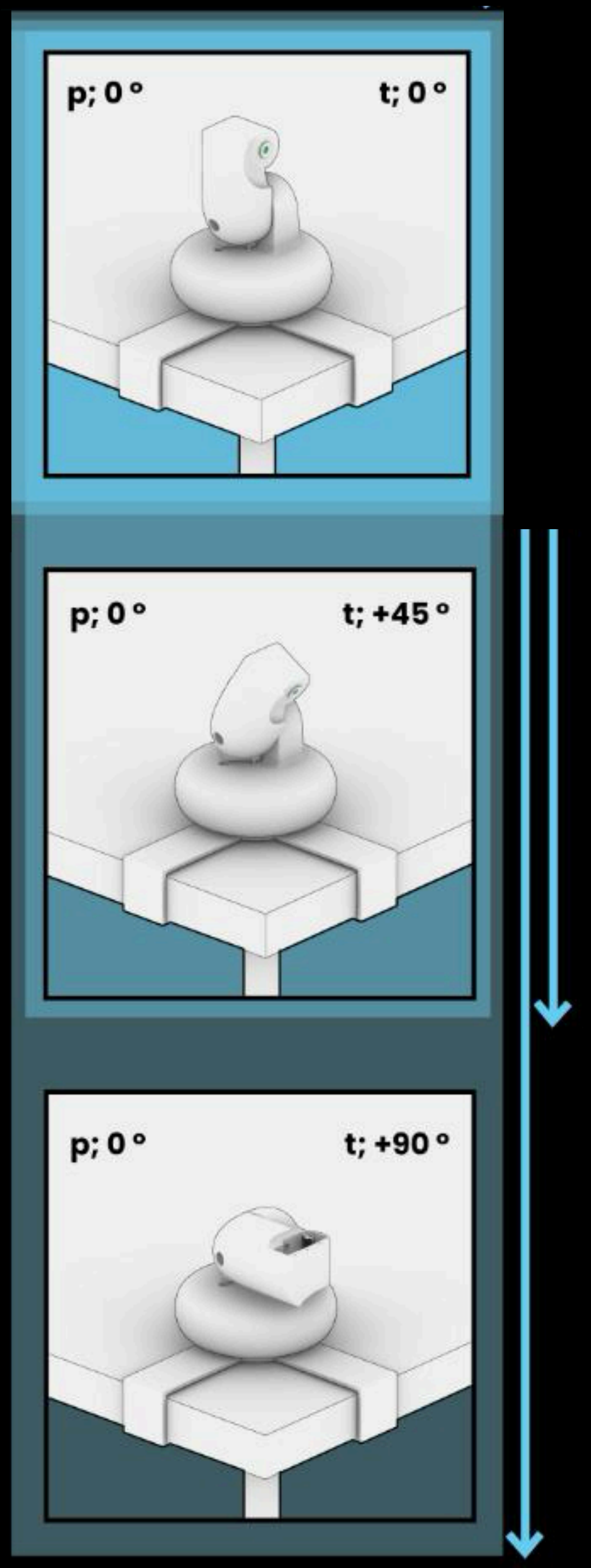


Gesture Library



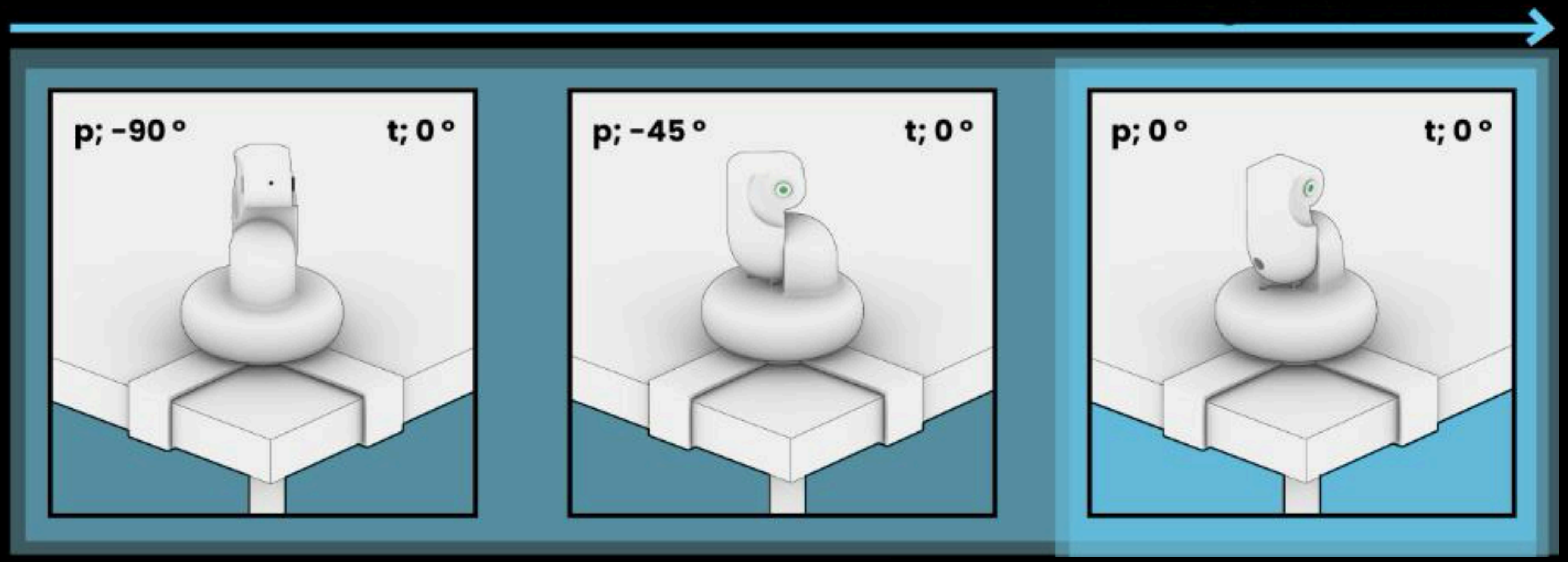
Yes/Positive / Greeting

Gesture Library



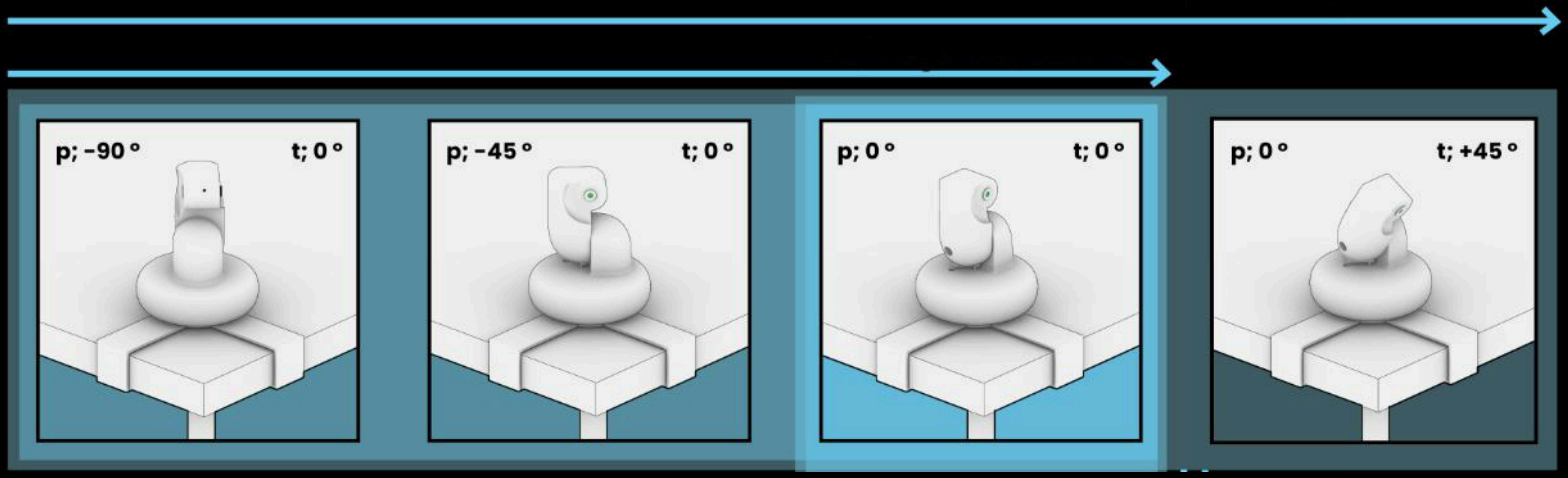
Thank you/ Your Welcome / Bye

Gesture Library



No/ Negetive/ Cant do

Gesture Library



I dont know/ I’m not sure

SYSTEM ARCHITECTURE / DATA FLOW

From a spoken phrase to a servo movement.

PointoBot is a four-stage pipeline that turns ambient lab speech into a physical pointing gesture. Each stage runs on a different host and communicates over UDP or shared files.

01 PIPELINE

STAGE 01 — MAC · MIC

Listen

voice.py

Continuous mic stream. Volume-based VAD captures speech segments and ignores ambient lab noise.

PYAUDIO · VAD 1500 · PRE-BUFFER 0.5S

→

STAGE 02 — MAC · LOCAL

Transcribe

faster-whisper

Captured audio is decoded locally to text. Empty or sub-threshold clips are dropped before routing.

WHISPER BASE.EN · INT8 · CPU

→

STAGE 03 — CLOUD · LLM

Route

Gemini 2.5 Flash

A structured JSON router decides: ignore chatter, fire a gesture, look up a known target, or recall a marked point.

JSON SCHEMA · TEMP 0.1 · SYSTEM PROMPT

→

STAGE 04 — PI · SERVOS

Move

pointo.py

Pi receives UDP, smooths target deltas, drives pan/tilt servos. Watchdog auto-homes after 10s of idle.

UDP :5005 · SMOOTHING 0.4 · CALIBRATED PWM

DECISION / ACCENT · FILE / ARTIFACT · STAGE / MODULE · CHANNEL / BOUNDARY

SYSTEM ARCHITECTURE / DATA FLOW

From a spoken phrase to a servo movement.

PointoBot is a four-stage pipeline that turns ambient lab speech into a physical pointing gesture. Each stage runs on a different host and communicates over UDP or shared files.

02 ROUTING LOGIC

Decision Engine

What did they mean?

A single Gemini call returns one of four action classes. Output is constrained by Pydantic schema so the router can switch deterministically.

→ RouterDecision { action }

IGNORE	**Drop the utterance.**	Background chatter, partial sentences, off-topic talk. No movement.
ANIM_*	**Fire a gesture animation.**	YES · NO · THANKS · IDK → UDP direct to Pi. Locks motion until done.
GH_TARGET	**Hand off to Grasshopper.**	Write target ID to target.txt. GH solves IK + pushes pan/tilt to Pi.
CUSTOM	**Recall a marked point.**	Look up pan/tilt in memory.json. UDP coordinates direct to Pi.

DECISION / ACCENT · FILE / ARTIFACT · STAGE / MODULE · CHANNEL / BOUNDARY

SYSTEM ARCHITECTURE / DATA FLOW

From a spoken phrase to a servo movement.

PointoBot is a four-stage pipeline that turns ambient lab speech into a physical pointing gesture. Each stage runs on a different host and communicates over UDP or shared files.

03 DATA STORES & CHANNELS

{ }

memory.json

Custom marks

Named pan/tilt pairs captured by mark.py. Loaded at boot into the LLM system prompt as recallable targets.

.txt

target.txt

GH handoff

A single integer ID written by voice.py and polled by the Grasshopper script every 10 seconds.

⇆

UDP :5005

Live channel

Both gestures (ANIM_*) and pan/tilt coordinate strings (pan,tilt) flow over the same socket to the Pi.

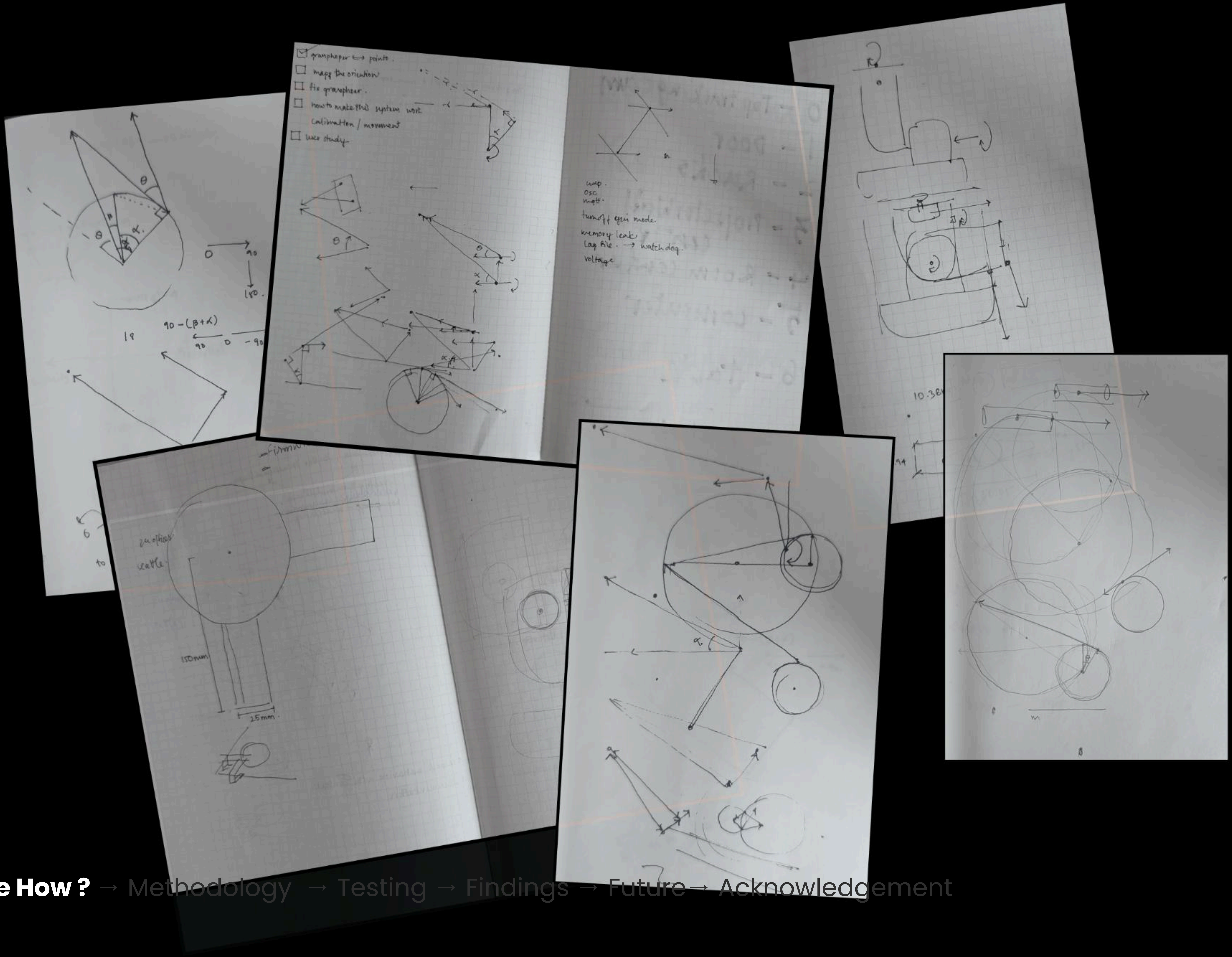
.csv

marks.csv

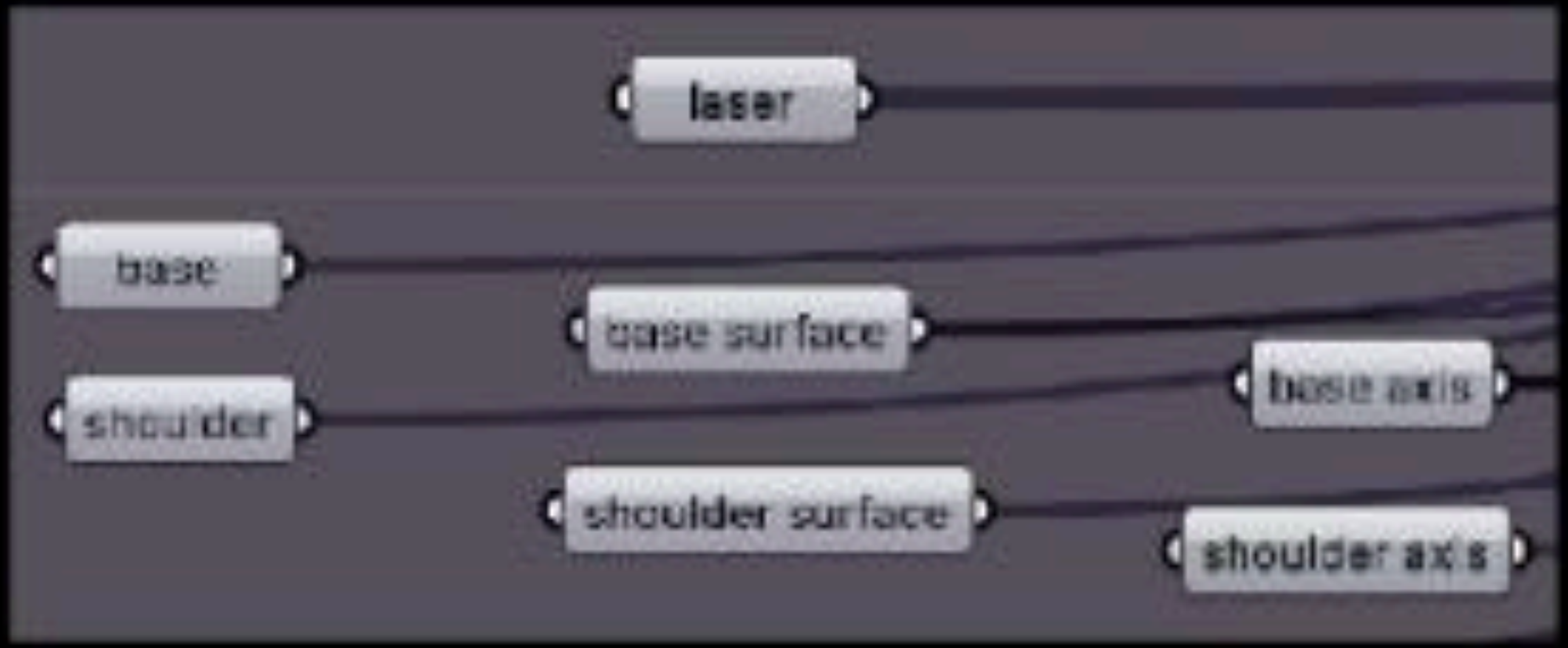
Calibration log

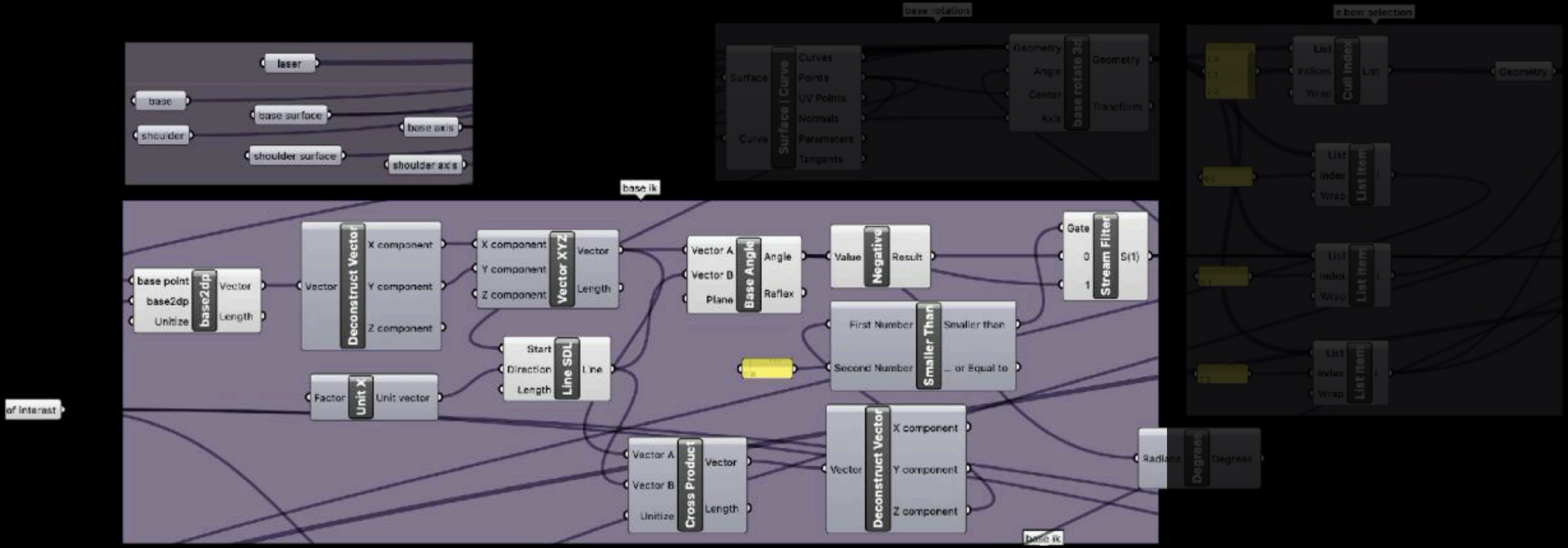
Append-only history of every marked point — name, pan, tilt, timestamp. Source of truth for memory.json.

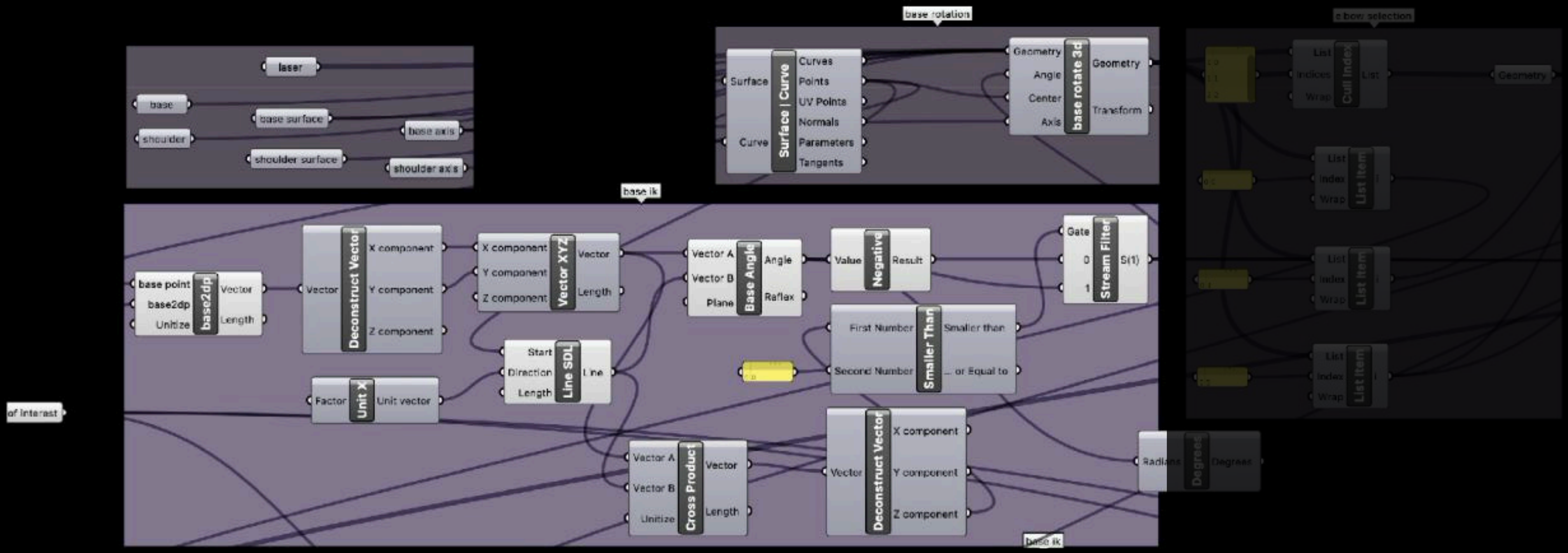
DECISION / ACCENT · FILE / ARTIFACT · STAGE / MODULE · CHANNEL / BOUNDARY



graphpaper to points.
mapp the orientation
fix graphpaper.
how to make the system work.
calibration / movement
user study.
turn off gui mode.
memory leak
log file. → watch dog.
voltage
150mm
25mm


laser
base
base surface
base axis
shoulder
shoulder surface
shoulder axis


laser
base
base surface
base axis
shoulder
shoulder surface
shoulder axis
base rotation
Surface
Curve
Surface | Curve
Curves
Points
UV Points
Normals
Parameters
Tangents
Geometry
Angle
Center
Axis
base rotate 3d
Transform
List
Indices
Wrap
Cull Index
Geometry
base ik
base point
base2dp
Unitize
base2dp
Vector
Length
Vector
Deconstruct Vector
X component
Y component
Z component
X component
Y component
Z component
Vector XYZ
Vector
Length
Vector A
Vector B
Plane
Base Angle
Angle
Reflex
Value
Negative
Result
Gate
0
1
Stream Filter
S(1)
First Number
Second Number
Smaller Than
Smaller than
... or Equal to
Start
Direction
Length
Line SDL
Line
Factor
Unit X
Unit vector
of interest
Vector A
Vector B
Unitize
Cross Product
Vector
Length
Vector
Deconstruct Vector
X component
Y component
Z component
Radians
Degrees
base ik



base rotation
laser
base
base surface
base axis
shoulder
shoulder surface
shoulder axis
Surface
Curve
Surface | Curve
Curves
Points
UV Points
Normals
Parameters
Tangents
Geometry
Angle
Center
Axis
base rotate 3d
Geometry
Transform
base ik
base point
base2dp
Unitize
base2dp
Vector
Length
Vector
Deconstruct Vector
X component
Y component
Z component
X component
Y component
Z component
Vector XYZ
Vector
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Vector A
Vector B
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Base Angle
Angle
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Value
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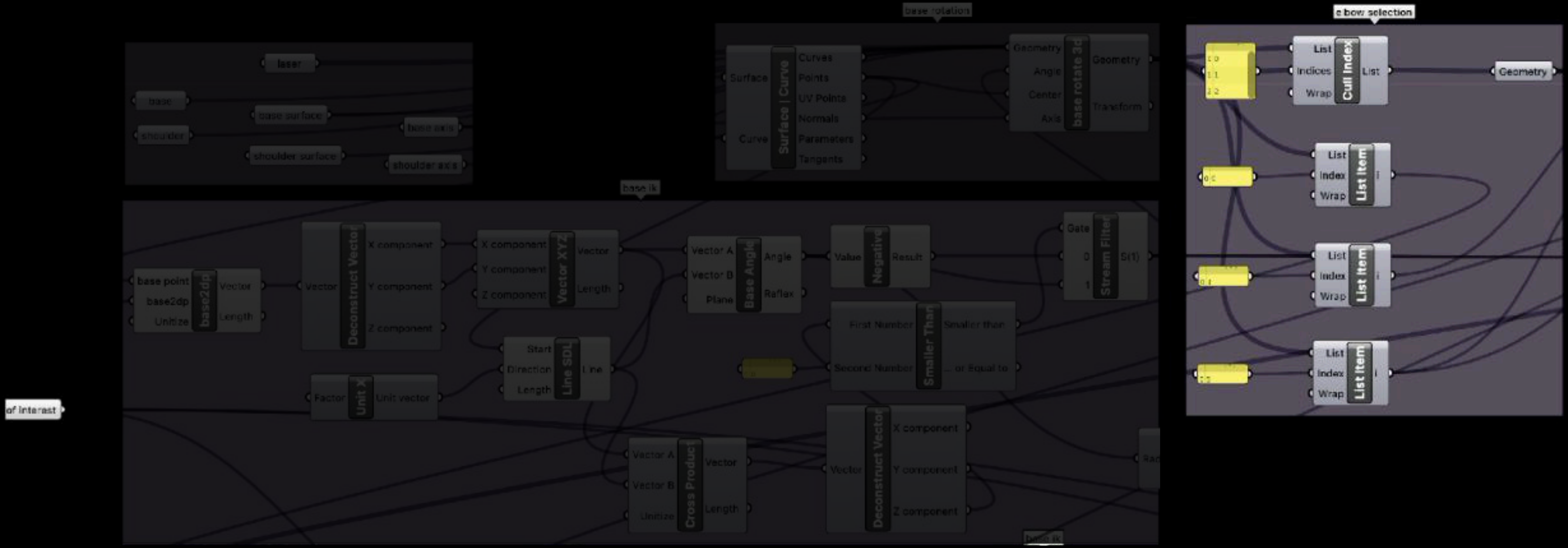
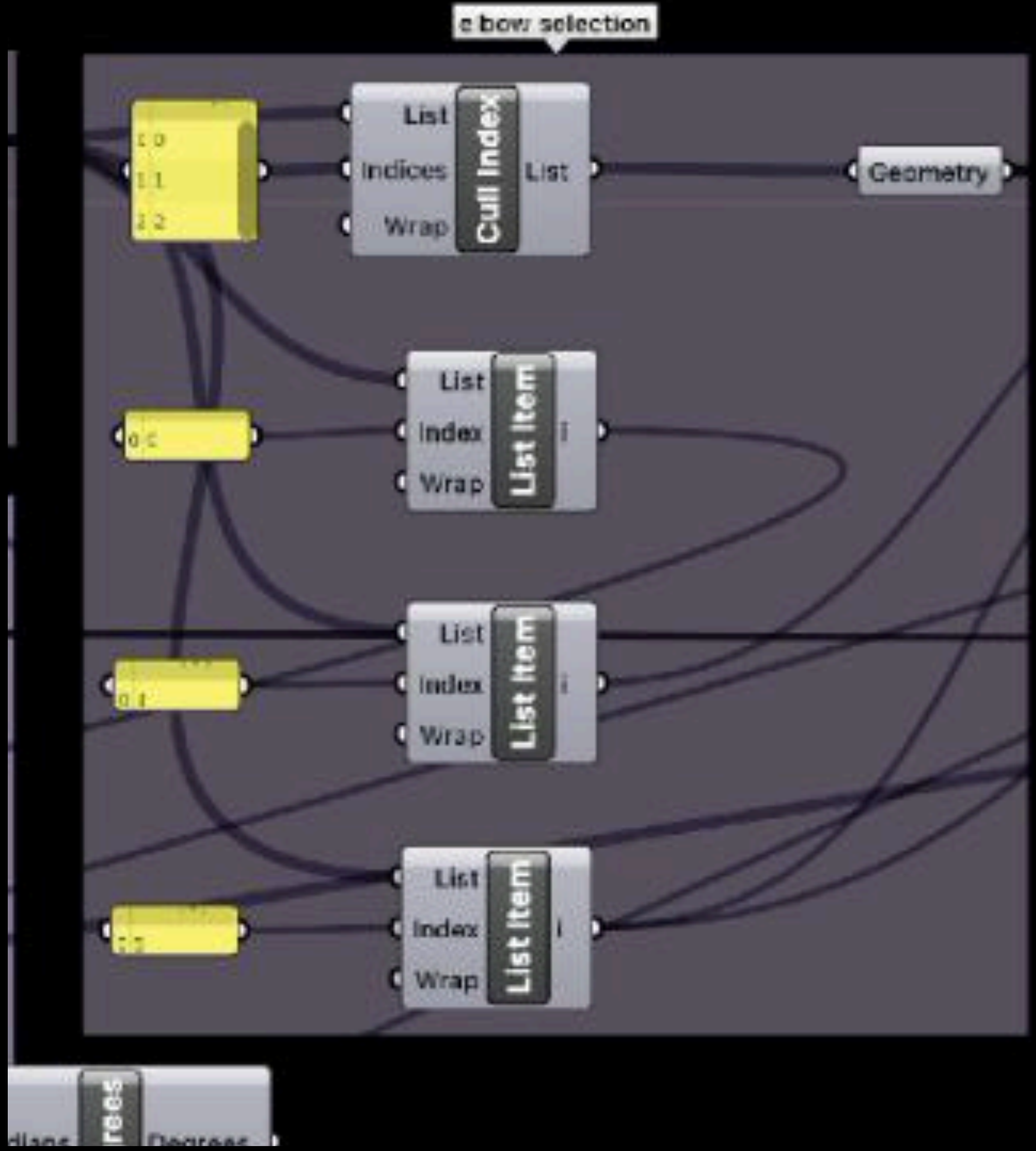
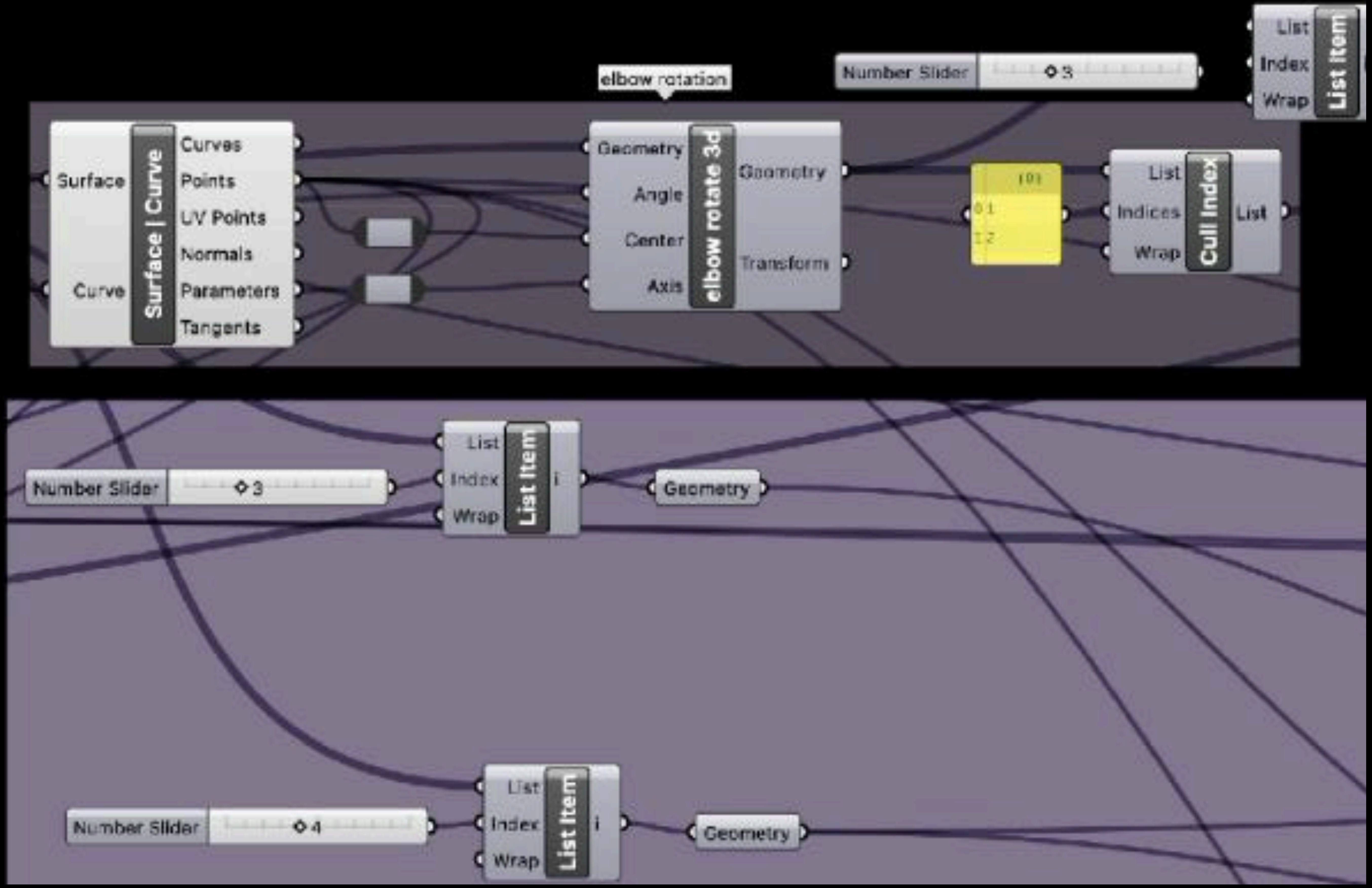
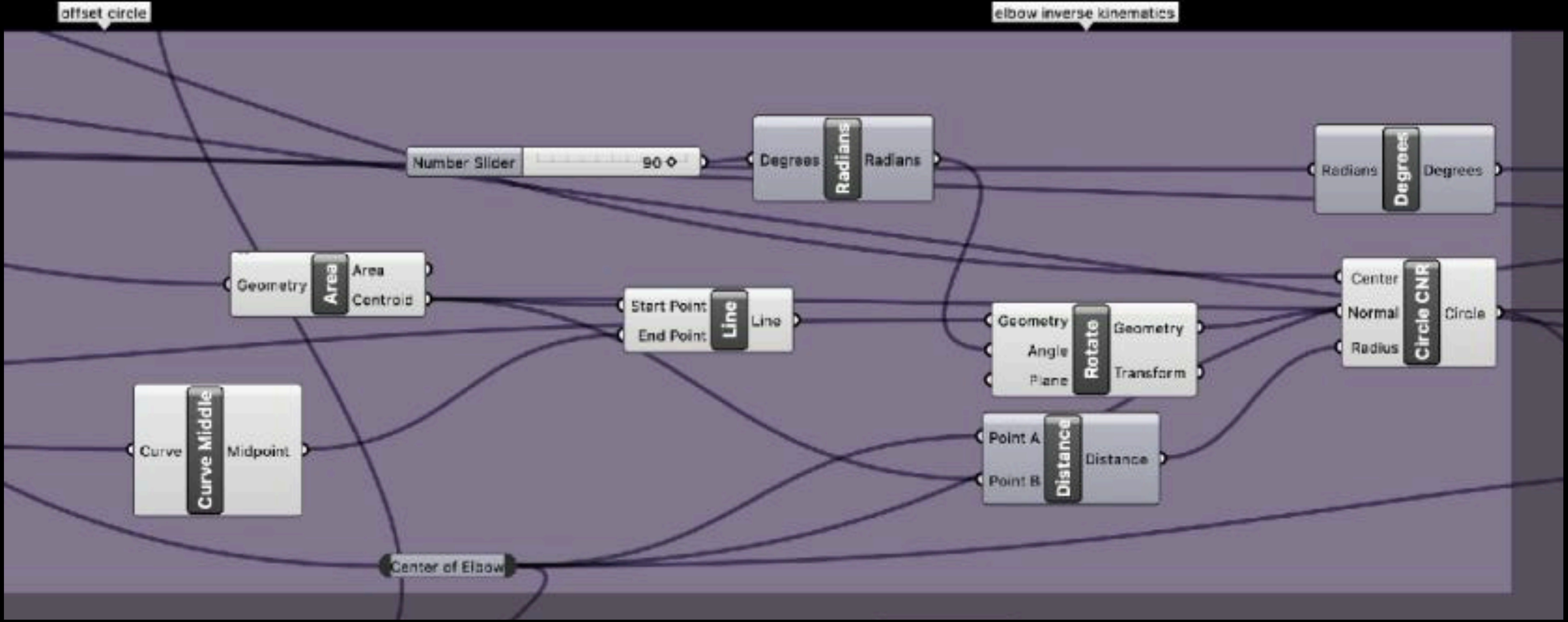
Start
Direction
Length
Line SDL
Line
Factor
Unit X
Unit vector
of interest
Vector A
Vector B
Unitize
Cross Product
Vector
Length
Vector
Deconstruct Vector
X component
Y component
Z component
Radians
Degrees
Degrees
base ik

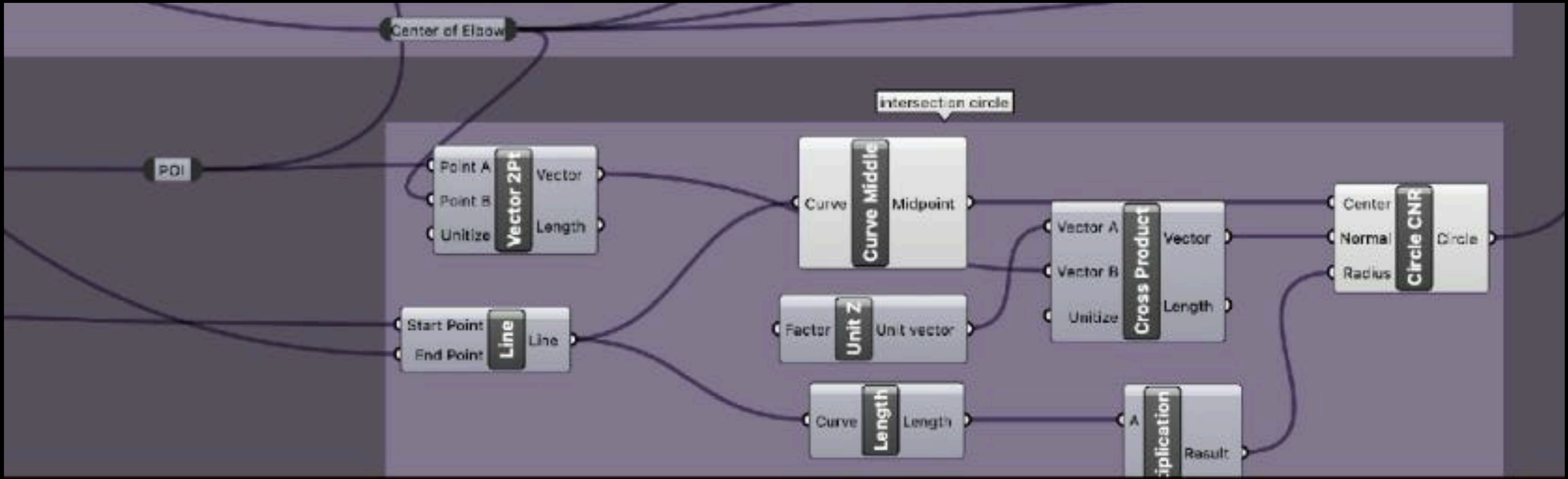


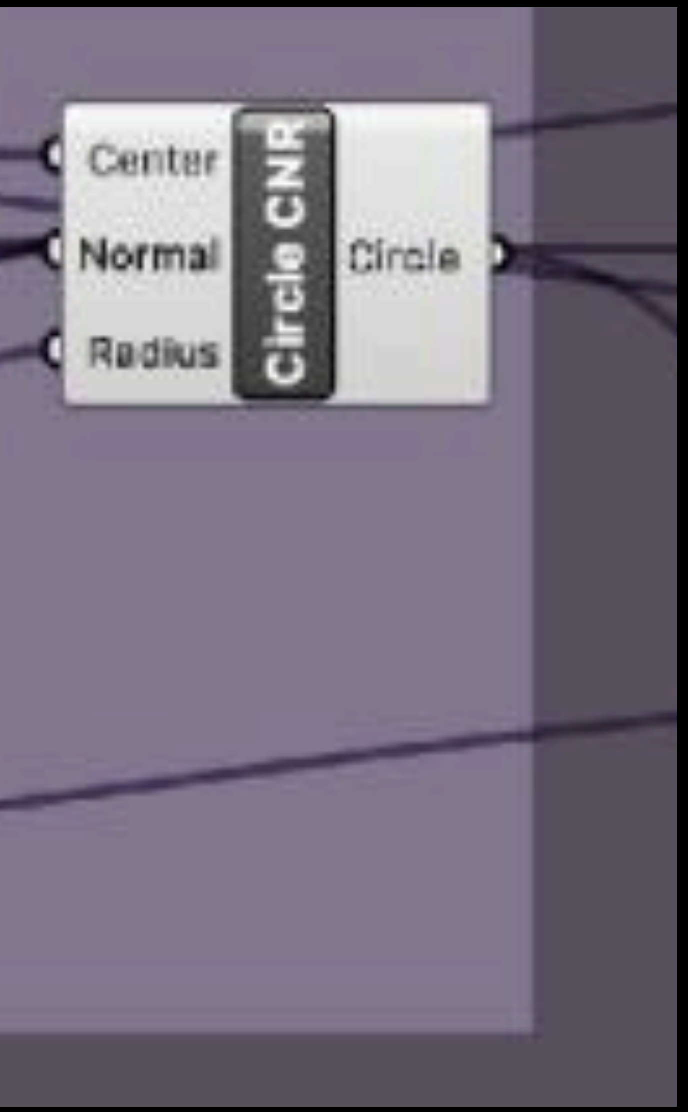
laser
base
base surface
base axis
shoulder
shoulder surface
shoulder axis
base rotation
Surface
Curve
Surface | Curve
Curves
Points
UV Points
Normals
Parameters
Tangents
Geometry
Angle
Center
Axis
base rotate 3d
Transform
elbow selection
List
Indices
Wrap
Cull Index
Index
List Item
base ik
base point
base2dp
Unitize
Vector
Length
Deconstruct Vector
X component
Y component
Z component
Vector XYZ
Vector A
Vector B
Plane
Base Angle
Reflex
Value
Negative
Result
Gate
Stream Filter
First Number
Second Number
Smaller Than
Smaller than
... or Equal to
Start
Direction
Line SDL
Line
Factor
Unit X
Unit vector
Cross Product
of interest

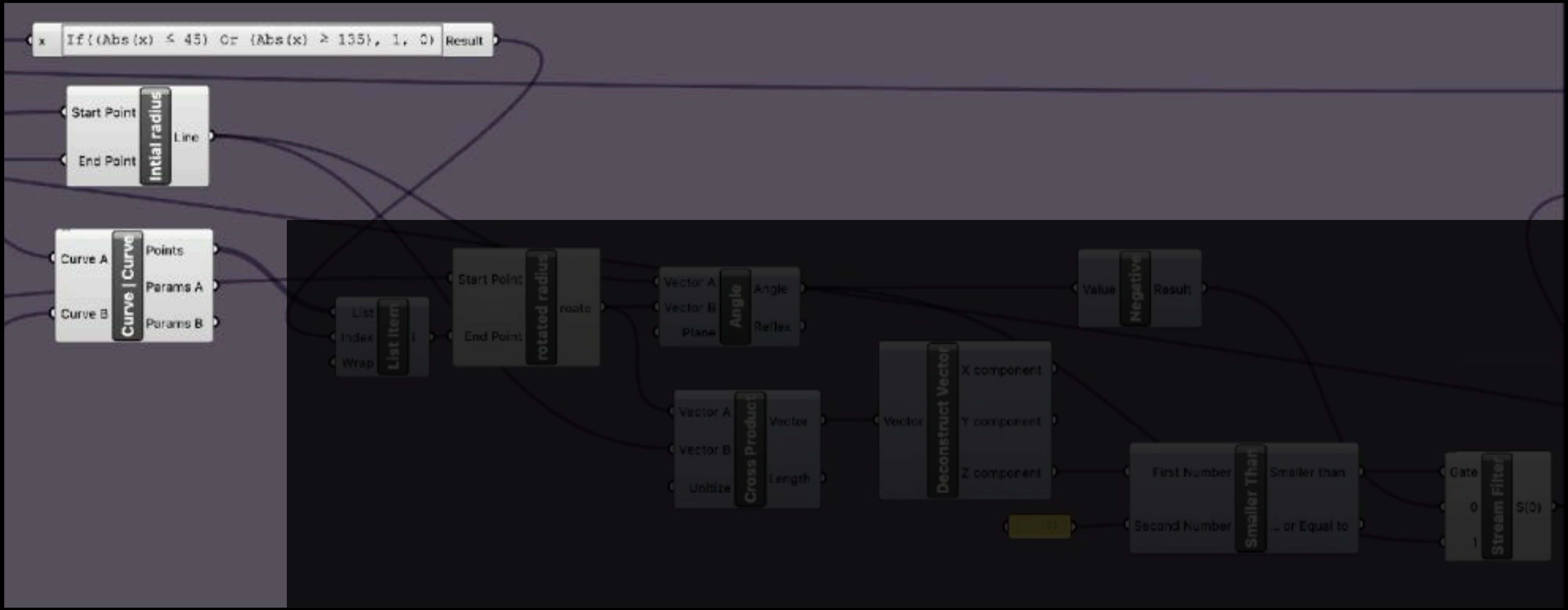

e bow selection
List
Indices
Wrap
Cull Index
List
Geometry
List
Index
Wrap
List Item
List
Index
Wrap
List Item
List
Index
Wrap
List Item

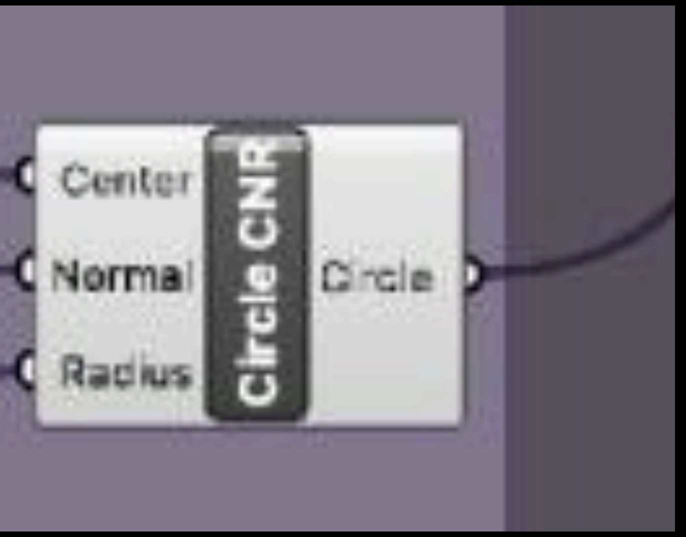

elbow rotation
Number Slider
3
List
Index
Wrap
List Item
Surface
Curve
Surface | Curve
Curves
Points
UV Points
Normals
Parameters
Tangents
Geometry
Angle
Center
Axis
elbow rotate 3d
Geometry
Transform
List
Indices
Wrap
Cull Index
List
Number Slider
3
List
Index
Wrap
List Item
Geometry
Number Slider
4
List
Index
Wrap
List Item
Geometry

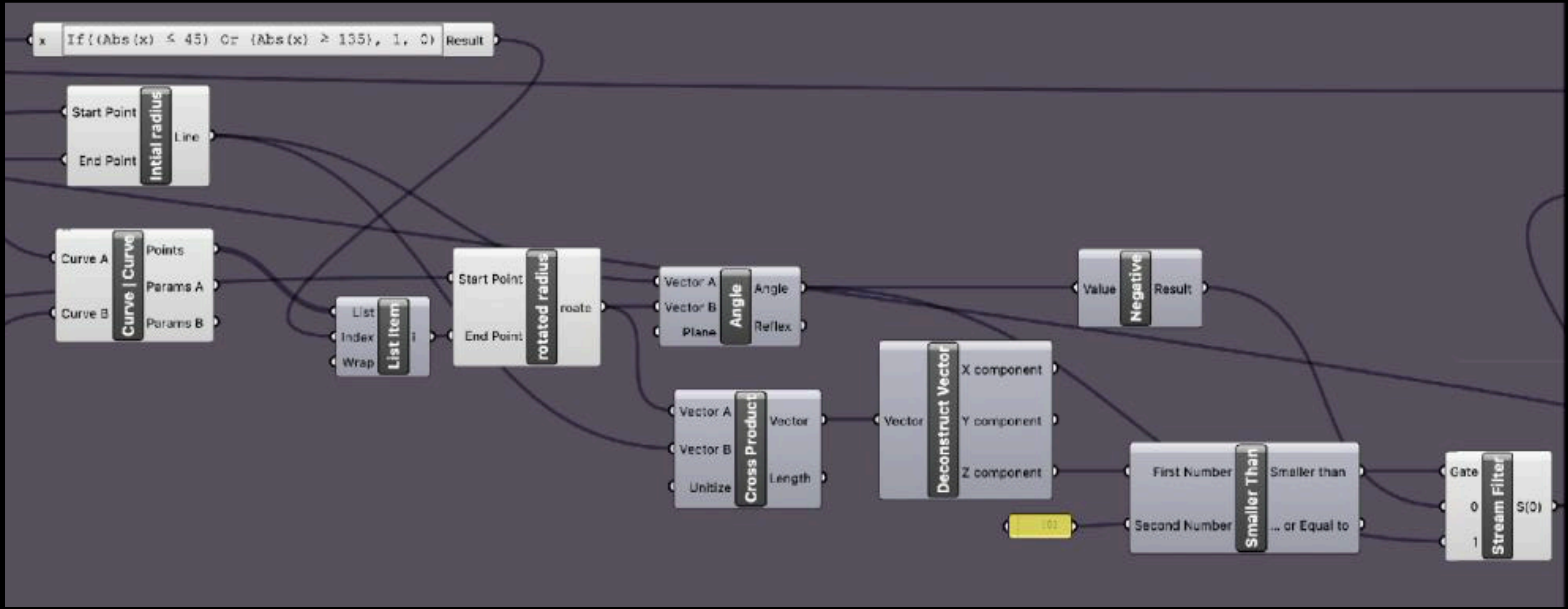

offset circle
elbow inverse kinematics
Number Slider
90
Degrees
Radians
Radians
Radians
Degrees
Degrees
Geometry
Area
Area
Centroid
Start Point
End Point
Line
Line
Geometry
Angle
Plane
Rotate
Geometry
Transform
Center
Normal
Radius
Circle CNR
Circle
Curve
Curve Middle
Midpoint
Point A
Point B
Distance
Distance
Center of Elbow


Center of Elbow
intersection circle
POI
Point A
Point B
Unitize
Vector 2Pt
Vector
Length
Curve
Curve Middle
Midpoint
Vector A
Vector B
Unitize
Cross Product
Vector
Length
Center
Normal
Radius
Circle CNR
Circle
Start Point
End Point
Line
Line
Factor
Unit Z
Unit vector
Curve
Length
Length
A
Result


Center
Normal
Radius
Circle CNR
Circle

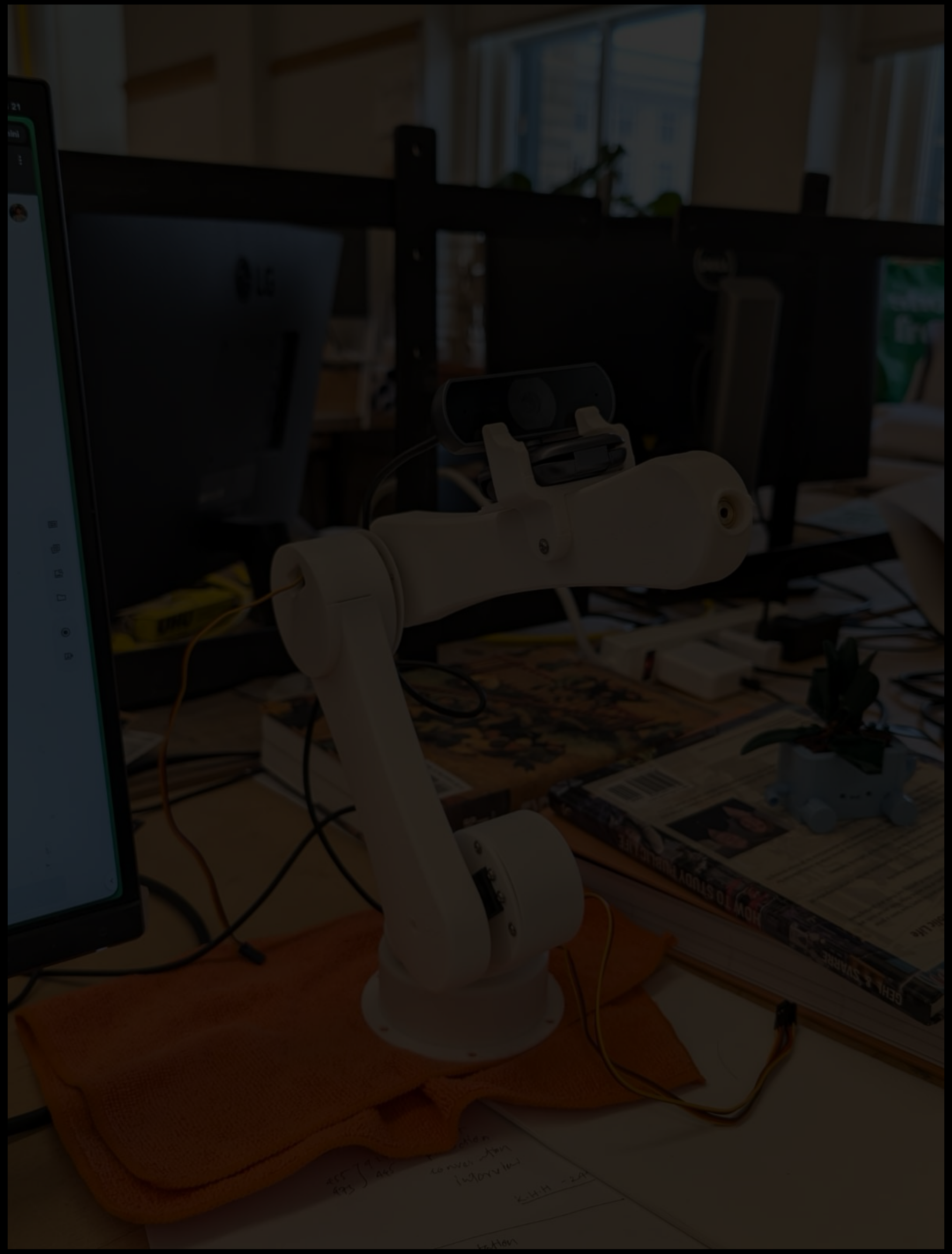

x
If((Abs(x) ≤ 45) Or (Abs(x) ≥ 135), 1, 0)
Result
Start Point
End Point
Intial radius
Line
Curve A
Curve B
Curve | Curve
Points
Params A
Params B
List
Index
Wrap
List Item
Start Point
End Point
rotated radius
roate
Vector A
Vector B
Plane
Angle
Reflex
Value
Negative
Result
Vector A
Vector B
Unitize
Cross Product
Vector
Length
Vector
Deconstruct Vector
X component
Y component
Z component
First Number
Second Number
Smaller Than
Smaller than
... or Equal to
Gate
0
1
Stream Filter
S(0)

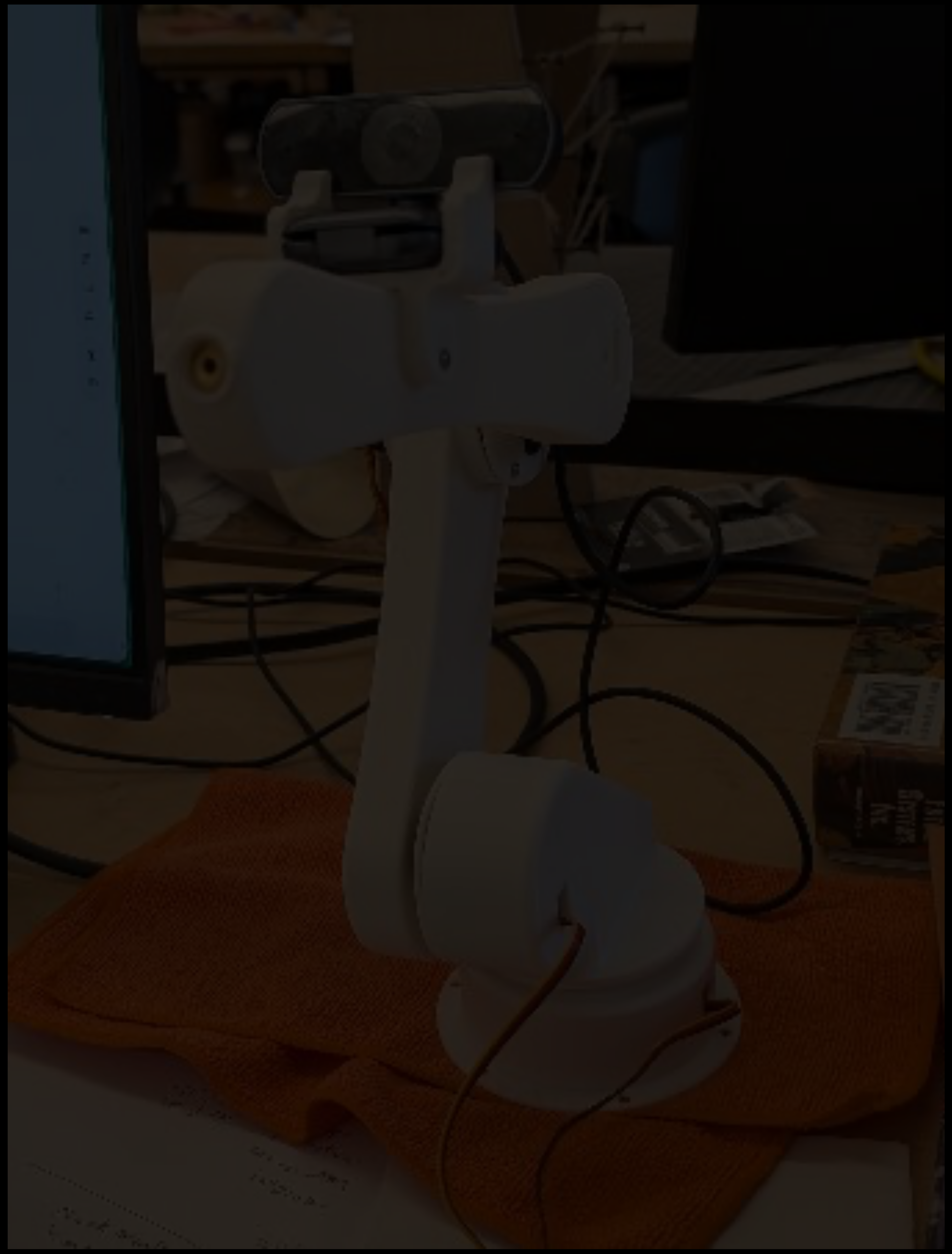

Center
Normal
Radius
Circle CNR
Circle

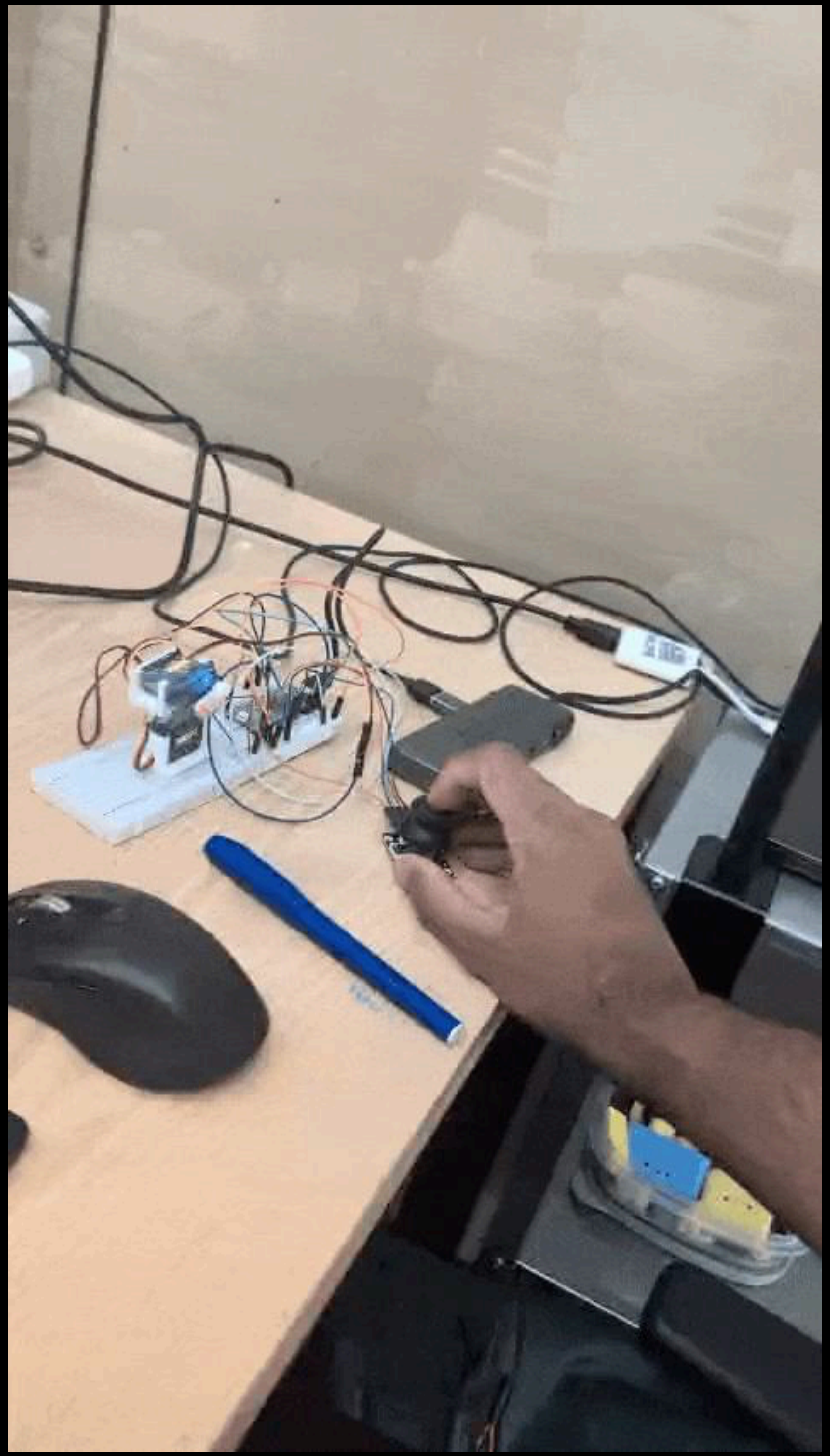

x
If((Abs(x) ≤ 45) Or (Abs(x) ≥ 135), 1, 0)
Result
Start Point
End Point
Intial radius
Line
Curve A
Curve B
Curve | Curve
Points
Params A
Params B
List
Index
Wrap
List Item
Start Point
End Point
rotated radius
roate
Vector A
Vector B
Plane
Angle
Angle
Reflex
Vector A
Vector B
Unitize
Cross Product
Vector
Length
Vector
Deconstruct Vector
X component
Y component
Z component
Value
Negative
Result
First Number
Second Number
Smaller Than
Smaller than
... or Equal to
Gate
0
1
Stream Filter
S(0)

pointo v1 {Dec 2025}

pointo v0 {Sept 2025}







prototyping w VLMs v1 {Sept 2025}